Sample Paper
FINALTERM EXAMINATION
Fall 2022
CS301P - Data Structures Practical

Time: 90 min
Marks: 40
Section A: Objective (20 marks)
Section B: Subjective (20 marks)
Mandatory : **20 MCQs**, **4 Long** (5 marks each)

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Solved by Qalb-e-Momin

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Question No: 1 (Marks: 01) - Please choose the correct option

What is the main advantage of linked lists over arrays?
a. Dynamic size
b. Faster indexing
c. Better cache performance
d. Constant time insertion and deletion

Question No: 2 (Marks: 01) - Please choose the correct option

How do you find the middle node in a linked list?
a. By dividing the size of the linked list by 2
b. By starting from the head node and traversing half of the list
c. By starting from the tail node and traversing half of the list
d. By randomly selecting a node

Question No: 3 (Marks: 01) - Please choose the correct option

What is the time complexity of accessing an element in a stack implemented using an array?
a. O(1)
b. O(n)
c. O(log n)
d. O(n log n)

Question No: 4 (Marks: 01) - Please choose the correct option

What is an underflow condition in a stack?
a. When the stack is empty and an element is popped
b. When the stack is full and an element is pushed
c. When the stack is not full but an element is not pushed
d. When the stack is not empty but an element is not popped

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Question No: 5 (Marks: 01) - Please choose the correct option

What is the use of a queue in computer science?
a. To store and retrieve elements in a sequential manner
b. To store and retrieve elements in a random manner
c. To store and retrieve elements in a fixed order
d. To store and retrieve elements in a reversed order

Question No: 6 (Marks: 01) - Please choose the correct option

If the elements "A", "B", "C" and "D" are placed in a queue and are deleted one at a time, in what order will they be removed?
a) ABCD // (Queue: FIFO - First In First Out)
b) DCBA
c) DCAB
d) ABDC

Question No: 7 (Marks: 01) - Please choose the correct option

What is a node in a tree that has no children called?
a. Leaf node
b. Root node
c. Parent node
d. Child node

Question No: 8 (Marks: 01) - Please choose the correct option

What is a degenerate tree?
a. A binary tree that is always balanced
b. A binary tree that is never balanced
c. A binary tree where all nodes have only one child
d. A binary tree where all nodes have two children

//A **degenerate tree** (also known as a **pathological tree**) is a special case of a binary tree in which **each parent node has only one child**. This makes the tree behave **like a linked list**, resulting in:

Poor performance (e.g., search time becomes **O(n)** instead of **O(log n)**).

No branching, just a straight line of nodes.

Question No: 9 (Marks: 01) - Please choose the correct option

Which of the following traversal outputs the data in sorted order in a BST?

a. Inorder

b. Pre order

c. Post order

d. Level Order

Question No: 10 (Marks: 01) - Please choose the correct option

While inserting the elements 71, 65, 84, 69, 67, 83 in an empty binary search tree (BST) in the sequence shown, the element in the lowest level is:
a. 65
b. 67
c. 69
d. 83

Question No: 11 (Marks: 01) - Please choose the correct option

How are reference variables created in programming?
a. By declaring a variable with the reference type
b. By declaring a variable with the primitive type
c. By calling a constructor
d. By initializing a variable with a value

Question No: 12 (Marks: 01) - Please choose the correct option

What happens when a constant variable is assigned a new value in programming?
a. The value of the constant variable is changed
b. The constant variable retains its old value
c. A compile-time error occurs
d. A runtime error occurs

Question No: 13 (Marks: 01) - Please choose the correct option

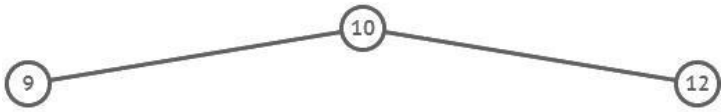
If values 20,30,25 are used to build AVL tree then which type of rotation can balance the AVL tree?

a) Double left-right
b) Double right-left
c) Single right
d) Single left

Question No: 14 (Marks: 01) - Please choose the correct option

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In the given BST, the balance factor of root node is:



a) 0
b) -1
c) 2
d) -2

Question No: 15 (Marks: 01) - Please choose the correct option

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The ASCII value for character 'C is:

a) 65
b) 66
c) 67
d) 68

Question No: 16 (Marks: 01) - Please choose the correct option

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In Huffman coding, data in a tree always occur?

a) roots
b) leaves
c) left sub trees
d) right sub trees

Question No: 17 (Marks: 01) - Please choose the correct option

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Consider the max heap represented by the following array. Which of the following is the parent of node 19?

55,45,35,25,30,31,32,23,19

a) 25
b) 35
c) 45
d) 55

Question No: 18 (Marks: 01) - Please choose the correct option

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Consider a min heap, represented by the following array:

15,16,18,19,17,20

After deleting the root node. Which of the following is the updated min heap?

a) 16,17,18,19,20
b) 16,17,19,18,20
c) 16,20,17,19,18
d) 16,20,18,19,20

Question No: 19 (Marks: 01) - Please choose the correct option

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In a max-heap, element with the greatest key is always in the which node?
a) Leaf node
b) First node of left sub tree
c) root node
d) First node of right sub tree

Question No: 20 (Marks: 01) - Please choose the correct option

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If we call the union function(2,3) and then (2,7), the name of set will be,
a) 2
b) 3
c) 7
d) 0

Question No: 21 (Marks: 01) - Please choose the correct option

---

Suppose in a hashing technique, we have the key and hash function given below:

Key = 20

Hash function = key / 10

At which array index, the key will be stored?

a) 2
b) 3
c) 7
d) 0

Question No: 22 (Marks: 01) - Please choose the correct option

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If Hash function = key %10, so far which of the pair of values, the collision take place?

a) 12 and 2
b) 14 and 15
c) 11 and 22
d) 1 and 10

Question No: 23 (Marks: 01) - Please choose the correct option

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Here is an array of ten integers 15 13 18 19 11 17 10 12 16 14
Array after the FIRST iteration of the main loop in **selection sort** algorithm will be (sorting from smallest to largest)

a) 10	13	18	19	11	17	15	12	16	14
b) 10	13	18	19	11	17	12	15	16	14
c) 13	10	18	19	11	17	12	15	16	14
d) 13	18	10	19	11	17	12	15	16	14

Question No: 24 (Marks: 01) - Please choose the correct option

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Which one of the following is not $N\log_2(N)$ algorithm?
a) Selection sort
b) Merge sort
c) Quick sort
d) Heap sort

//

🔍 Time Complexities:

Algorithm	Average/Best Time Complexity
Selection Sort	$O(n^2)$ ❌ not N log N
Merge Sort	O(n log n) ✅
Quick Sort	O(n log n) on average ✅
Heap Sort	O(n log n) ✅

//

Question No: 25 (Marks: 05)

Consider the given C++ code and write C++ code for the following :

1. Take 5 odd numbers as input from the user
2. Declare an object of Stack class
3. Push the numbers in Stack class object

```
#include <iostream>
#include “Node.h”
#include “Stack.h”
using namespace std;
int main(){

//Write code here

}
```

Answer

```
Stack s; // Step 2: Declare stack object

    int count = 0;
    int number;

    // Step 1: Take 5 odd numbers as input
    cout << "Enter 5 odd numbers:" << endl;
    while (count < 5) {
        cin >> number;
        if (number % 2 != 0) {
            s.push(number); // Step 3: Push into stack
            count++;
        } else {
            cout << "Please enter an **odd** number." << endl;
        }
    }

    return 0;
```

```cpp
#include <iostream>
#include "Node.h"
#include "Stack.h"
using namespace std;

int main(){

    Stack s; // Step 2: Declare stack object

    int count = 0;
    int number;

    // Step 1: Take 5 odd numbers as input
    cout << "Enter 5 odd numbers:" << endl;
    while (count < 5) {
        cin >> number;
        if (number % 2 != 0) {
            s.push(number); // Step 3: Push into stack
            count++;
        } else {
            cout << "Please enter an **odd** number." << endl;
        }
    }

    return 0;
}
```

Question No: 26 (Marks: 05)

Show the result of inserting values 4,3,6,7,11,5,8 into an initially empty AVL tree.
(Draw only final AVL tree. Steps are not required)

Answer



Question No: 27 (Marks: 05)

Suppose, we have been given the following array.

11	8	4	13	6
0	1	2	3	4

You are required to sort the above data in ascending order using Bubble sort.

Note: You have to show the steps of 1st iteration only.

Answer

..

Step 1: Compare index 0 and 1 → 11 > 8 → **Swap**

[11, 8, 4, 13, 6] → [8, 11, 4, 13, 6]

Step 2: Compare index 1 and 2 → 11 > 4 → **Swap**

[8, 11, 4, 13, 6] → [8, 4, 11, 13, 6]

Step 3: Compare index 2 and 3 → 11 < 13 → **No swap**

[8, 4, 11, 13, 6]

Step 4: Compare index 3 and 4 → 13 > 6 → **Swap**

[8, 4, 11, 13, 6] → [8, 4, 11, 6, 13]

✅ **Array after 1st Iteration:**

[8, 4, 11, 6, 13]

Question No: 28 (Marks: 05)

Consider the given next () function of List class. Find out the error(s) in the given code and write correct statement(s).

```
string next ()
{
if (currentNode == NULL)
      return false+currentNode;
else {
currentNode = currentNode->void.getNext() ;
return true;
 }
return true;
```

Answer

```
string next ()
{
if (currentNode == NULL)
      return"" ;
else {
currentNode = currentNode->getNext() ;
          return "true";
}
          return "true";
```

Question No: 29 (Marks: 05)

Write the output of given code?

```
Queue a;
Stack b;
a = new Queue();
b = new Stack();
b.push(11);
b.push(7);
b.push(9);
a.enqueue(s.pop());
a.enqueue(10);
cout<<b.pop();
a.enqueue(7);
cout<< a.dequeue();
s.push(a.dequeue());
cout<<b.pop();
cout<<b.pop();
```

Answer

791011

How?

Queue a;

Stack b;

a = new Queue(); // Redundant line (object already created)

b = new Stack(); // Redundant line (object already created)

b.push(11); // Stack: [11]

b.push(7); // Stack: [7, 11]

b.push(9); // Stack: [9, 7, 11]

a.enqueue(b.pop()); // pop() = 9 → Queue: [9], Stack: [7, 11]

a.enqueue(10); // Queue: [9, 10]

cout << b.pop(); // Output: 7 → Stack: [11]

a.enqueue(7); // Queue: [9, 10, 7]

cout << a.dequeue(); // Output: 9 → Queue: [10, 7]

b.push(a.dequeue()); // dequeue() = 10 → Stack: [10, 11], Queue: [7]

cout << b.pop(); // Output: 10 → Stack: [11]

cout << b.pop(); // Output: 11 → Stack: []

❖ **Final Answer:** 791011

Question No: 30 (Marks: 05)

Insert the values 2, 3, 9, 11 into the following table using hash function (3x + 5)mod9. You have to apply linear probing technique to resolve collision.

0	
1	
2	
3	
4	
5	
6	

Answer

0	
1	
2	2
3	11
4	
5	3
6	9

How ?

Step-by-step Insertion:

1. Insert 2
   - h(2) = (3×2 + 5) mod 9 = (6 + 5) mod 9 = 11 mod 9 = 2
   - Index 2 is empty → insert 2 at index 2
2. Insert 3
   - h(3) = (3×3 + 5) mod 9 = (9 + 5) mod 9 = 14 mod 9 = 5
   - Index 5 is empty → insert 3 at index 5
3. Insert 9
   - h(9) = (3×9 + 5) mod 9 = (27 + 5) mod 9 = 32 mod 9 = 5
   - Index 5 is occupied → apply linear probing
   - Check index 6 → empty → insert 9 at index 6
4. Insert 11
   - h(11) = (3×11 + 5) mod 9 = (33 + 5) mod 9 = 38 mod 9 = 2
   - Index 2 is occupied → linear probing
   - Check index 3 → empty → insert 11 at index 3